

UNIT: 5: Introduction to Electromagnetism



=> Topic:-

- ✓ ✓ -> Gradient
 - ✓ ✓ -> divergence
 - ✓ ✓ -> curl
- } their physical significance
- ✓ -> Divergence and curl of electrostatic and static magnetic fields,
 - > Faraday's law.
 - > Equation of continuity
 - > Displacement current
 - ✓ -> Maxwell's equations
 - ✓ -> Electromagnetic wave propagation in free space. Flow of energy and Poynting vector.

DEL (∇) operator:-

- > The divergence, gradient and curl all involve partial derivatives there is a notation employ that can express the operation succinctly (clearly).
- > Let the DEL operator be defined in cartesian coordinates by the formal expression.

$$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$$

- > This is a vector differential operator that ~~exists~~ acts on functions and vector fields through the technical

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right), \text{ the gradient}$$

$$\nabla F = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right), F = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot (f_x, f_y, f_z)$$

$$= \frac{\partial f_x}{\partial x}, \frac{\partial f_y}{\partial y}, \frac{\partial f_z}{\partial z}, \text{ the divergence}$$

$$\nabla \times F = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \times F$$

$$= \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_x & f_y & f_z \end{bmatrix} \text{ the curl}$$

Q) Determine the divergence of a vector field in two dimensions $F(x, y) = 6x^2 \hat{i} + 4y \hat{j}$.

Soln:-

$$F(x, y) = 6x^2 \hat{i} + 4y \hat{j}$$

$$\nabla F(x, y) = \frac{\partial f_1}{\partial x}, \frac{\partial f_2}{\partial y}$$

$$= 12x + 4$$

Ex:- $F(x, y, z) = x^2 \hat{i} + 2z \hat{j} - y \hat{k}$

$$\nabla F(x, y, z) = 2x + 0 + 0$$

$$= 2x$$

(Q) find curl $F(x, y, z) = y^3 \hat{i} + xy \hat{j} - z \hat{k}$

$$\nabla \times F(x, y, z) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^3 & xy & -z \end{vmatrix}$$

$$= \hat{i} \left(\frac{\partial(-z)}{\partial y} - \frac{\partial(xy)}{\partial z} \right) + \hat{j} \left(\frac{\partial(-z)}{\partial x} - \frac{\partial(y^3)}{\partial z} \right) + \hat{k} \left(\frac{\partial(xy)}{\partial x} - \frac{\partial(y^3)}{\partial y} \right)$$

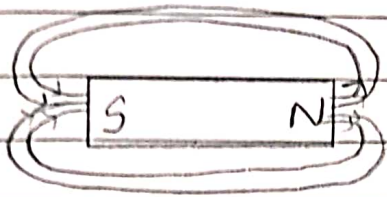
$$= 0 \hat{i} + 0 \hat{j} + (y - 3y^2) \hat{k}$$

$$= (y - 3y^2) \hat{k}$$

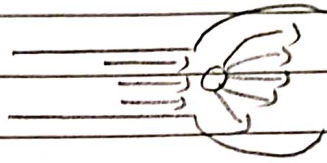
Physical Significance Divergence, curl and gradient.

=> Divergences:-

→ Divergence is the net flow of field/liquid/ substance out of a unit volume.



$$\text{Div} = 0$$



$$\text{Div} \neq 0$$

=> CURL:-

→ curl gives the measure of angular velocity of a object.

→ If curl is zero, it means the object is not rotating.

→ If curl is not zero, its magnitude represents the speed of the object and its direction denotes the axis of rotation.



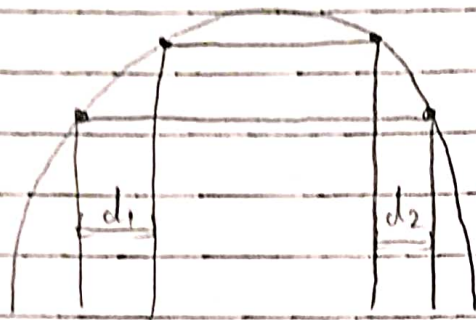
rotate = exist



No rotate $\neq 0$

⇒ Gradient:-

→ Gradient is the rate of change of a parameter P with respect to parameter Q .



Divergence and Curl of electrostatic and static magnetic fields:

⇒ Divergence ~~and curl~~ of magnetic field

→ The magnetic field due to volume current distribution given by biot-savart can be expressed as follows.

$$\vec{B} = \frac{\mu_0}{4\pi} \int_V \frac{\vec{J} \times \vec{r}}{r^2} dV \quad - (1)$$

→ To evaluate the divergence of the magnetic field the divergence is applied to both side of bio-savart.

$$\nabla \cdot \vec{B} = \frac{\mu_0}{4\pi} \int_V \nabla \cdot \left(\vec{J} \times \frac{\vec{r}}{r^2} \right) dV \quad - (2)$$

→ Applying the vector product rule the term within the integral is simplified.

$$\vec{\nabla} \cdot \vec{B} = \frac{\mu_0}{4\pi} \int_V \left(\frac{\vec{r}}{r^2} \cdot (\vec{\nabla} \times \vec{J}) - \vec{J} \cdot (\vec{\nabla} \times \frac{\vec{r}}{r^2}) \right) dV \quad \text{--- (3)}$$

→ The first term involving the curl of the current density function is zero. Since the current density is independent of the field coordinate.

→ Using vector analysis the second term in the above eqn also reduces to zero. Hence the divergence of magnetic field zero.

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \text{--- (4)}$$

→ The zero divergence of the magnetic field is valid for any field with ~~res~~ irrespective of whether the field is static or time dependent.

⇒ Curl of magnetic field:-

→ To evaluate the curl of magnetic field the curl is applied to both side of Bio-Savart law.

$$\vec{\nabla} \times \vec{B} = \frac{\mu_0}{4\pi} \int_V \vec{\nabla} \times \left(\vec{J} \times \frac{\vec{r}}{r^2} \right) dV \quad \text{--- (5)}$$

→ Again by applying vector analysis the eqn is simplified.

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} \quad - (6)$$

→ The curl of magnetic field equals the vacuum permeability multiplied by the current density.

→ The same result is obtained by applying Stokes's theorem to the integral form of Ampere's law.

→

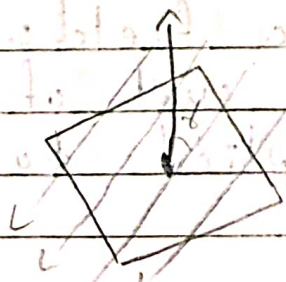
$$\oint_S (\vec{\nabla} \times \vec{B}) \cdot d\vec{A} = \mu_0 \oint_S \vec{J} \cdot d\vec{A} \quad - (7)$$

→ Since the above relation holds for any close loop the integrands are equal this equation is called the differential form of Ampere's law.

⇒ Curl and divergence of electrostatic field:

→ Divergence :-

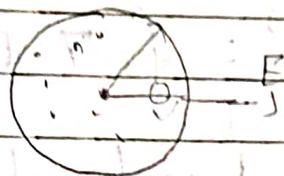
→ A measure of the number of field lines passing perpendicularly through the surface S is called Flux.



$$\phi_E = \int_S \vec{E} \cdot d\vec{a} \quad - (8)$$

→ Thus, for a closed surface integration it will give a measure of the total sources and sink inside the close surface.

→ Let us calculate the simplest example of a point charge and uses the simplest surface of a spherical surface central point at the charge itself.



$$\phi_E = \oint_S \vec{E} \cdot d\vec{a}$$

$$= \oint_S \frac{q}{4\pi\epsilon_0 r^2} \hat{r} \cdot \hat{r} r^2 \sin\theta d\theta d\phi$$

$$= \frac{q}{4\pi\epsilon_0} (2)(2\pi) = \frac{q}{\epsilon_0}$$

→ For a collection of point charges by a the superposition principle

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4 + \dots$$

→ The flux through closed gaussian surface

$$\phi_E = \oint_S \vec{E} \cdot d\vec{a} = \oint_S \sum_{i=1}^n \dots d\vec{a}$$

$$= \sum_{i=1}^n \frac{q_i}{\epsilon_0}$$

or

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{\text{enclosed}}}{\epsilon_0} \rightarrow \text{Gauss' law}$$

→ The previous Gauss law in integral form can be written also as differential form using the divergence theorem.

$$\oint_S \vec{E} \cdot d\vec{a} = \int_V \nabla \cdot \vec{E} d\tau \quad - (A)$$

and the enclosed charge written as

$$Q_{\text{enc}} = \int_V \rho(\vec{r}) d\tau \quad - (B)$$

combining (A) and (B)

$$\int_V \nabla \cdot \vec{E} d\tau = \int_V \frac{\rho}{\epsilon_0} d\tau$$

Since the volume of integration can be taken arbitrary then the integrands must be equal.

$$\boxed{\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}}$$

→ curl of a electric field :-

→ The electric field of a point charge at the origin is given by

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \quad \text{--- (1)}$$

looking at the radially directed electric field lines produced by the point charge. we can deduced their curl of e.f is zero.

→ Let's calculate the curl of general electric field produced by a charged distribution.

$$\vec{\nabla} \times \vec{E} = \vec{\nabla} \times \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{r^2} \hat{r} d\tau$$

⇒ Maxwell's Equation: Physical Meaning:-

- Maxwell's Equations are composed of four equations with each one describes one phenomenon respectively.
- Maxwell didn't invent all these equations, but rather he combined the four equations made by Gauss, Faraday and Ampere.
- But Maxwell added one piece of information into Ampere's law (the 4th eqⁿ) - Displacement Current, which makes the equation complete.
- ① Gauss's law for static electric fields.
- ② Gauss's law for static magnetic fields.
- ③ Faraday's law which says a changing magnetic field produces an electric field.

④ Ampere - Maxwell's law which says a changing electric field produces a magnetic field.

→ The combination of eqⁿ ③ and ④ can explain electromagnetic wave which can propagate on its own.

$$\nabla \cdot \mathbf{D} = \rho_v$$

$$\nabla \cdot \mathbf{B} = 0$$

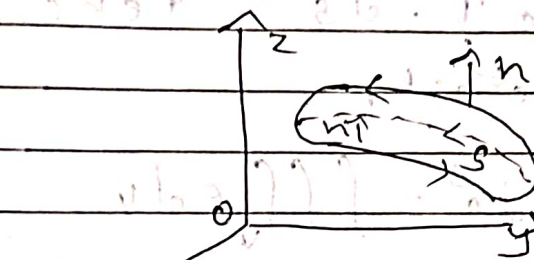
$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}$$

Stokes' Theorem Formula:-

→ The Stokes' theorem states that "the surface integral of the curl of a function over a surface bounded by a closed surface is equal to the line integral of the particular vector function around that surface".

→ Stokes' theorem formula:-



$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$$

Divergence theorem statement:-

→ The divergence theorem states that the surface integral of the normal component of a vector point function "F" over a closed surface "S" is equal to the volume integral of the divergence of F taken over the volume "V" enclosed by the surface S. Thus, the divergence theorem is symbolically denoted as:

$$\iiint_V \nabla \cdot \vec{F} \cdot dV = \iint_S \vec{F} \cdot \vec{n} \cdot dS.$$

Proof of Maxwell first equation:-

→ mathematically gauss law can be expressed as

$$\oint_S (\vec{E} \cdot d\vec{s}) = \frac{Q_{enc}}{\epsilon_0}$$

$$\oint_S \epsilon_0 \vec{E} \cdot d\vec{s} = Q_{enc}$$

$$\oint_S \vec{D} \cdot d\vec{s} = Q_{enc}$$

Density

$$[Q = \rho V]$$

Volume density

$$\iiint_S \vec{D} \cdot d\vec{s} = \iiint_V \rho \, dv$$

$$[Q_{enc} = \iiint_V \rho \, dv]$$

By divergence theorem we,

$$\boxed{\oint_S \vec{D} \cdot d\vec{s} = \iiint_V \vec{\nabla} \cdot \vec{D} \, dv} \quad \text{Formula}$$

$$\iiint_V \vec{\nabla} \cdot \vec{D} \, dv = \iiint_V \rho \, dv$$

$$\boxed{\vec{\nabla} \cdot \vec{D} = \rho}$$

Proof of Maxwell second eqn:-

→ By gauss law for magnetic field.

$$\iiint_S \vec{B} \cdot d\vec{s} = 0$$

applying divergence theorem

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$$\boxed{\iint_S \vec{B} \cdot d\vec{s} = \iiint_V (\vec{\nabla} \cdot \vec{B}) d\vec{v}} \quad \text{Formula ...}$$

$$\iiint_V (\vec{B} \cdot \vec{\nabla}) d\vec{v} = \iiint_V 0 d\vec{v}$$

$$\boxed{\vec{\nabla} \cdot \vec{B} = 0}$$

Proof of Maxwell Third equation:-
→ From Faraday's law,

$$\text{emf} = -\frac{d\phi}{dt} \quad \text{--- (1)}$$

Here the scalar magnetic flux can be replaced by

$$\phi = \iint_S \vec{B} \cdot d\vec{s} \quad \text{--- (2)}$$

substituting eqn (2) in eqn (1)

$$\text{emf} = -\frac{d}{dt} \iint_S \vec{B} \cdot d\vec{s} \quad \text{--- (3)}$$

which is a partial eqn given by

$$\text{emf} = -\iint_S \frac{d}{dt} \vec{B} \cdot d\vec{s} \quad \text{--- (4)}$$

The alternative electromotive force induced in a coil is basically a closed path so

$$\text{emf} = \oint \vec{E} \cdot d\vec{l} \quad \text{--- (5)}$$

↳ closed surface integral

From eqn (4) & (5)

$$\oint \vec{E} \cdot d\vec{l} = - \iint_S \frac{\partial \vec{B}}{\partial t} \cdot \frac{d\vec{S}}{\partial t} \quad \text{--- (6)}$$

line integral surface integral

From Stokes's theorem on C.H.S

$$\oint \vec{E} \cdot d\vec{l} = \oint (\vec{\nabla} \times \vec{E}) \cdot d\vec{S} \quad \text{--- (7)}$$

putting eqn - (6) in eqn - (7)

$$\oint_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{S} = - \oint_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

$$\boxed{\therefore \vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}}$$

Proof of Maxwell's fourth eqn.

→ By Ampere's law,

$$\oint \vec{B} \cdot d\vec{l} = \mu I$$

$$\oint \frac{\vec{B}}{\mu} \cdot d\vec{l} = I$$

$$\oint \vec{H} \cdot d\vec{l} = I$$

$$\oint_C \vec{H} \cdot d\vec{l} = \iint_S \vec{J} \cdot d\vec{S}$$

$$I = \iint_S \vec{J} \cdot d\vec{S}$$

Applying Stokes's theorem

$$\iiint_V (\vec{\nabla} \times \vec{H}) \cdot d\vec{s} = \iiint_V \vec{J} \cdot d\vec{s} \left[\oint_S \vec{H} \cdot d\vec{l} = \iiint_V (\vec{\nabla} \times \vec{H}) \cdot d\vec{s} \right]$$

$$\boxed{\vec{\nabla} \times \vec{H} = \vec{J}} \quad \text{--- (1)}$$

applying divergence & divergence on both side

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \cdot \vec{J}$$

$$0 = \vec{\nabla} \cdot \vec{J}$$

($\because$ Divergence of curl is zero)

$$\boxed{\therefore \vec{\nabla} \cdot \vec{J} = 0} \quad \text{--- (2)}$$

But by eqⁿ of continuity

$$\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t} \quad \text{--- (3)}$$

as there was conflict so,

maxwell modified the eqⁿ $[\vec{\nabla} \times \vec{H} = \vec{J} + \vec{J}_d]$ from eqⁿ (1)

$$\vec{\nabla} \times \vec{H} = \vec{J} + \vec{J}_d \quad \text{--- (A)}$$

Now applying divergence on both side in eqⁿ (A)

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = (\vec{\nabla} \cdot \vec{J}) + (\vec{\nabla} \cdot \vec{J}_d)$$

$\therefore$ divergence of curl is 0.

$$0 = \vec{\nabla} \cdot \vec{J} + \vec{\nabla} \cdot \vec{J}_d$$

$$\vec{\nabla} \cdot \vec{J} = -\vec{\nabla} \cdot \vec{J}_d \quad \text{by eqⁿ (3) or continuity}$$

$$\frac{\partial \rho}{\partial t} = \vec{\nabla} \cdot \vec{J}_d$$

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$$\boxed{\frac{\partial \phi}{\partial t} = \frac{\partial (\vec{\nabla} \cdot \vec{D})}{\partial t}} \quad \text{by Maxwell's first eqn}$$

$$\frac{\partial (\vec{\nabla} \cdot \vec{D})}{\partial t} = \vec{\nabla} \cdot \vec{J}_d$$

$$\vec{\nabla} \cdot \frac{\partial \vec{D}}{\partial t} = \vec{\nabla} \cdot \vec{J}_d$$

$$\boxed{\vec{J}_d = \frac{\partial \vec{D}}{\partial t}}$$

Faraday's law of Induction:-

$$\oint_C \vec{E} \cdot d\vec{x} = - \frac{d}{dt} \int_S (\vec{B} \cdot \vec{n}) dS$$

- Changing magnetic flux through a surface induces an electromotive force (emf) in any boundary path of that surface.
- A changing magnetic field induces a circulating electric field.
- The voltage accumulated around a closed circuit is proportional to the time rate of change of the magnetic flux it encloses.

⇒ Poynting vector:-

→ Poynting Vector 'S' is defined as the cross product of the vectors $\vec{E}$ and $\vec{H}$. The direction of power flow at any point is normal to both $\vec{E}$ and $\vec{H}$ vector. The unit is watts per metre square. (watt's/m²).

→ Poynting Theorem:-

from maxwell's 4th eqⁿ

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

taking dot product with ' $\vec{E}$ ' on both sides

$$\vec{E} \cdot (\nabla \times \vec{H}) = \vec{E} \cdot \vec{J} + \vec{E} \cdot \frac{\partial \vec{D}}{\partial t}$$

from vector identity

$$\nabla \cdot (\vec{E} \times \vec{H}) = \vec{H} \cdot (\nabla \times \vec{E}) - \vec{E} \cdot (\nabla \times \vec{H})$$

or

$$\vec{E} \cdot (\nabla \times \vec{H}) = \vec{H} \cdot (\nabla \times \vec{E}) - \nabla \cdot (\vec{E} \times \vec{H})$$

substituting value from maxwell's 3rd eqⁿ.

$$\vec{H} \cdot \left(-\frac{\partial \vec{B}}{\partial t} \right) - \nabla \cdot (\vec{E} \times \vec{H}) = \vec{E} \cdot \vec{J} + \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} \quad \text{--- (1)}$$

substituting $\vec{J} = \sigma \vec{E}$, $\vec{D} = \epsilon \vec{E}$ & $\vec{B} = \mu \vec{H}$ in eqⁿ-(1)

$$\nabla \cdot (\vec{E} \times \vec{H}) = -\sigma \vec{E}^2 - \epsilon \frac{\partial}{\partial t} (\vec{E} \cdot \vec{E}) - \mu \frac{\partial}{\partial t} (\vec{H} \cdot \vec{H})$$

$$E^2 = \vec{E} \cdot \vec{E}$$

$$\frac{\partial E^2}{\partial t} = E \frac{\partial E}{\partial t} + E \frac{\partial E}{\partial t}$$

$$= 2E \frac{\partial E}{\partial t}$$

$$\frac{1}{2} \frac{\partial E^2}{\partial t} = E \frac{\partial E}{\partial t}$$

$$\rightarrow \nabla \cdot (\vec{E} \times \vec{H}) = -\sigma E^2 - \epsilon \frac{1}{2} \frac{\partial E^2}{\partial t} - \mu \frac{1}{2} \frac{\partial H^2}{\partial t}$$

$$\nabla \cdot (\vec{E} \times \vec{H}) = -\sigma E^2 - \frac{\partial}{\partial t} \left[\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right]$$

→ Integrating through out the volume

$$\int_V \nabla \cdot (\vec{E} \times \vec{H}) dV = - \int_V \sigma E^2 dV - \int_V \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) dV$$

From divergence theorem

$$\int_S \vec{D} \cdot d\vec{s} = \int_V (\nabla \cdot \vec{D}) dV$$

$$\int_S (\vec{E} \times \vec{H}) \cdot d\vec{s} = \int_V \sigma E^2 dV - \int_V \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) dV$$

Total
Power

Power
Dissipation

Rate of
energy stored.

Electromagnetic wave propagation in free space:-
→ The wave is propagating in the z-direction and is polarised with the electric field parallel to x-axis and the magnetic field parallel to y-axis.

① For Electromagnetic wave, in Electric field.

$$\vec{E} = E_0 \cos(\omega t - \beta z) \hat{a}_x$$

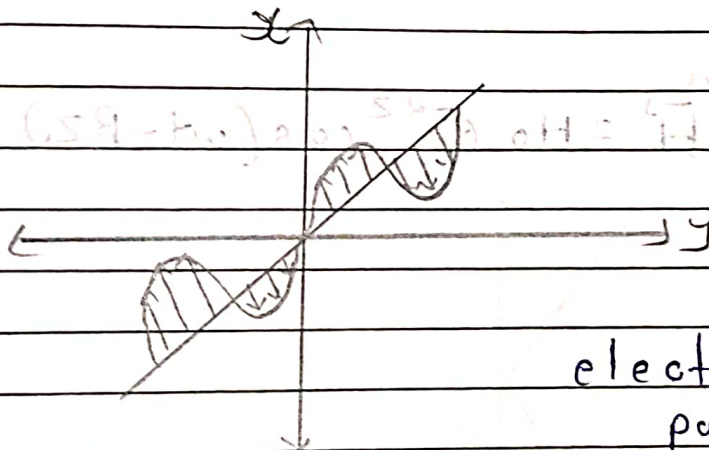
Here, $E_0 \rightarrow$ Amplitude

$\omega t \rightarrow$ frequency

$\beta \rightarrow$ phase difference

$z \rightarrow$ Direction of propagation

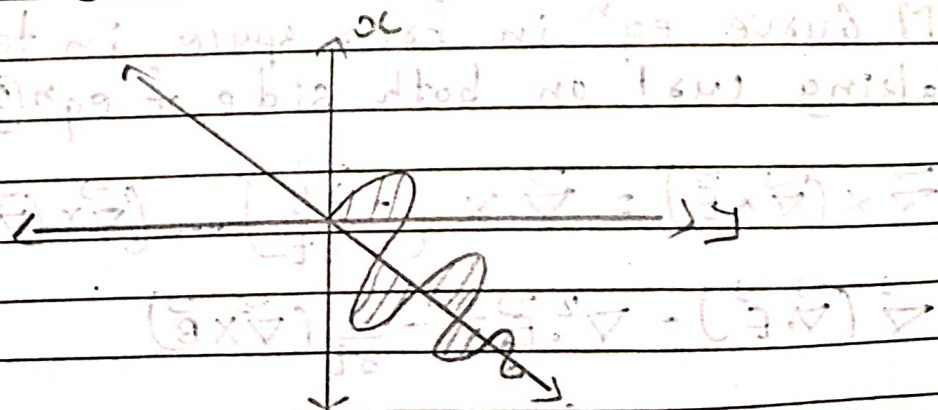
$\hat{a}_x \rightarrow$ " " Amplitude



electric field wave is parallel to x-axis.

② For in medium:-

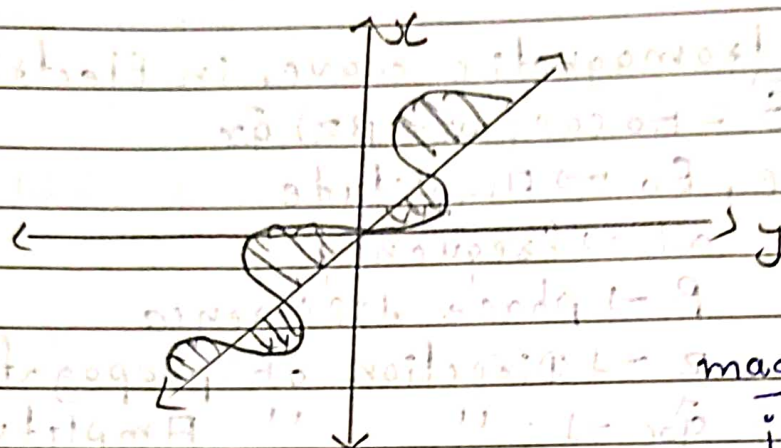
$$\vec{E} = E_0 e^{-\alpha z} \cos(\omega t - \beta z) \hat{a}_x$$



② For EM wave, magnetic field:

$$\vec{H} = H_0 \cos(\omega t - \beta z) \hat{a}_y \sim \begin{matrix} \text{direction of} \\ \text{amplitude} \end{matrix}$$

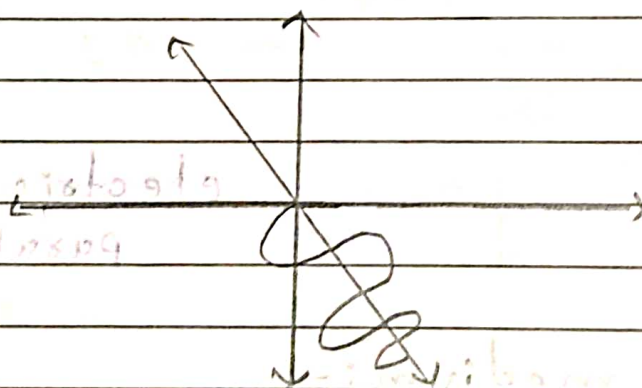
$\swarrow$ amplitude $\searrow$ frequency $\nearrow$ direction of propagation



magnetic field wave is parallel to y-axis

→ In medium

$$\vec{H} = H_0 e^{-\alpha z} \cos(\omega t - \beta z) \hat{a}_y$$



EM wave eqⁿ in free space in terms of $\vec{E}$.

→ Taking curl on both side of eqⁿ (3) of maxwell.

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \vec{\nabla} \times \left(-\frac{\partial \vec{B}}{\partial t} \right) \dots [\vec{\nabla} \times (\vec{\nabla} \cdot \vec{A}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A}]$$

$$\vec{\nabla} (\vec{\nabla} \cdot \vec{E}) - \nabla^2 \vec{E} = -\frac{\partial}{\partial t} (\vec{\nabla} \times \vec{B})$$

$$0 - \nabla^2 \vec{E} = -\frac{\partial}{\partial t} (\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t})$$

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$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad \text{--- (5)}$$

Similarly for B

$$\nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} \quad \text{--- (6)}$$